

ROUEN VALLEE DE SEINE AP, France

WMO: 070370

Lat: 49.3831N

Lon: 1.1817E

Elev: 156

StdP: 99.46

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-3.3	-8.3	1.9	-3.1	-6.4	2.2	-2.3	13.3	9.7	11.9	9.5	3.4	40	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.2	19.9	27.0	19.1	25.0	18.2	20.8	27.3	19.7	25.6	18.8	23.7	4.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.7	23.4	17.6	12.9	22.0	16.8	12.2	21.0	60.7	27.3	57.0	25.9	53.8	23.9	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.3	8.3	-7.4	33.3	2.2	2.7	-9.1	35.3	-10.4	36.9	-11.6	38.4	-13.3	40.4
			-8.0	23.0	2.3	1.0	-9.6	23.7	-11.0	24.3	-12.2	24.9	-13.9	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	4.1	4.8	7.2	9.7	12.8	15.8	18.0	17.8	15.0	11.7	7.5	4.6
	DBStd	5.97	3.85	3.61	3.40	3.36	3.29	3.17	3.09	3.07	2.82	3.18	3.30	3.77
	HDD10.0	760	183	148	97	44	9	1	0	0	1	19	87	171
	HDD18.3	2837	440	379	344	258	174	88	45	46	106	206	325	426
	CDD10.0	1051	1	2	12	36	97	176	247	243	152	72	12	2
	CDD18.3	87	0	0	0	0	3	14	33	31	6	1	0	0
	CDH23.3	836	0	0	0	4	29	144	317	285	53	3	0	0
	CDH26.7	231	0	0	0	0	2	36	91	93	9	0	0	0
Wind	WSAvg	4.1	4.8	4.7	4.6	4.2	4.0	3.7	3.6	3.4	3.7	4.2	4.3	4.6
Precipitation	PrecAvg	790	74	53	65	49	70	56	61	60	68	75	88	77
	PrecMax	1024	158	134	133	98	148	137	167	161	168	174	161	177
	PrecMin	479	16	6	8	6	17	2	11	2	17	11	19	5
	PrecStd	138	37	30	31	24	32	31	31	33	39	46	36	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.7	19.3	23.7	26.1	30.1	32.3	32.9	27.8	22.5	16.3	13.4
		MCWB	11.4	10.3	12.1	15.4	18.5	20.6	20.8	21.0	18.9	17.1	14.3	11.9
	2%	DB	11.8	12.1	16.4	20.6	23.9	27.2	29.1	29.1	24.7	19.5	14.5	12.3
		MCWB	10.6	9.7	11.3	14.2	16.9	19.4	19.7	19.9	17.9	15.9	12.9	11.0
	5%	DB	10.5	10.9	14.2	17.8	21.5	24.5	26.9	26.4	22.1	17.7	13.2	11.2
		MCWB	9.3	9.0	10.4	12.4	15.8	18.1	19.0	18.9	16.6	14.8	11.9	10.1
	10%	DB	9.3	9.8	12.3	15.6	19.2	22.2	24.7	24.1	20.2	16.3	12.2	10.1
		MCWB	8.3	8.3	9.7	11.2	14.5	16.8	18.2	17.8	15.6	14.1	11.0	9.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	11.2	13.1	16.6	19.6	21.9	22.0	22.3	20.2	17.9	15.1	12.5
		MCDB	12.6	12.9	16.7	22.5	25.0	28.4	30.2	29.8	25.6	21.1	15.8	13.0
	2%	WB	10.8	10.2	12.0	14.5	17.6	20.0	20.6	20.7	18.6	16.5	13.2	11.3
		MCDB	11.7	11.6	15.0	19.6	22.2	25.8	27.5	27.2	23.0	18.8	14.1	12.1
	5%	WB	9.5	9.4	11.1	12.9	16.1	18.7	19.5	19.5	17.4	15.4	12.2	10.3
		MCDB	10.3	10.6	13.6	16.8	20.4	23.7	25.3	25.3	21.0	17.2	13.1	11.2
	10%	WB	8.4	8.4	10.0	11.6	14.9	17.4	18.5	18.4	16.4	14.5	11.1	9.2
		MCDB	9.2	9.6	12.0	14.7	18.7	21.5	23.5	22.8	19.4	16.0	12.1	10.0
Mean Daily Temperature Range	5% DB	MDBR	4.9	6.0	7.5	9.2	9.3	9.9	10.2	10.2	9.3	7.4	5.6	4.9
		MCDBR	5.4	7.3	10.3	12.5	12.9	13.6	14.2	14.3	12.2	8.8	6.1	5.3
		MCWBR	4.8	5.4	6.4	7.2	7.3	7.2	6.5	6.5	6.2	5.7	4.8	4.8
	5% WB	MCDBR	5.2	6.4	8.6	11.1	11.5	12.3	12.9	12.7	10.3	7.9	5.7	5.2
		MCWBR	4.8	5.2	5.8	6.7	6.9	7.1	6.5	6.3	5.8	5.5	4.9	4.9
Clear-Sky Solar Irradiance	taub	0.321	0.338	0.372	0.393	0.400	0.400	0.397	0.390	0.368	0.355	0.329	0.314	
	taud	2.426	2.391	2.290	2.245	2.261	2.293	2.302	2.345	2.388	2.428	2.439	2.398	
	Ebn at Noon	729	804	830	851	859	860	856	846	827	773	711	678	
	Edh at Noon	64	84	109	126	129	126	123	113	98	80	63	58	
All-Sky Solar Radiation	RadAvg	0.89	1.64	2.78	4.30	5.03	5.63	5.37	4.58	3.54	2.06	1.07	0.76	
	RadStd	0.09	0.22	0.33	0.48	0.43	0.55	0.40	0.37	0.23	0.18	0.10	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page